

Figure 2: An interactive geometry program called Kig

introduction to a host of computer skills for the child.

You do not need an OLPC/XO to run Sugar. Sugar is available as a live CD (check their website). If you are running recent versions of Fedora, Sugar is available as a Desktop. Install it or run it as a live CD. Do not read the document first, but explore the wonderfully intuitive interface; if you get stuck you are probably crusty with age!

Senior classes

Geometry is fun when you take your swanky new instrument box to school the first time. But the fun quickly wears off, as the square you are trying to draw does not square off as expected! There is no substitute for practising geometry as your math teacher tells you to. However, there is *Kig* and *DrGeo* (www2.0fset.org/DrGeo) to liven things up. Both of these are interactive geometry programs with fairly advanced features and can be used as teaching aids in class as well.

Graphing and visualising are integral to advanced mathematics. *KAlgebra* and *KmPlot* are good entry-level tools to plot and study graphs. They could be used with advantage to teach linear programming, as well as linear and quadratic equations.

Marble is a good tool to learn geography—visualise day turning to night, measure distances, view thematic maps, and even how people imagined the world to be in 1689. It turns out they were not very far off the mark, which is impressive. You can play with a virtual version of the earth as you would with a marble. Perhaps that is how the name came about.

Step, courtesy the KDE project, is an interactive physics simulator for classical mechanics. You can design your own experiments or run the ones available. It can also make a great classroom teaching aid. I wish I had it when I was learning physics.

For chemistry buffs there is *Kalzium* from KDE and *gElemental* from GNOME to learn the periodic table. The GNOME Chemistry Utils has a suite of small programs useful in chemistry.

KStars is an accurate desktop planetarium. *Celestia* (shatters.net/celestia) is a 3D space simulator that can provide the young astronomer hours of stimulation and learning.

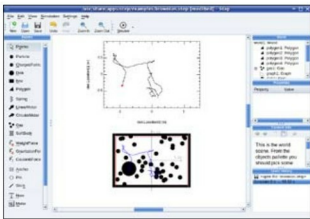


Figure 3: An interactive physics simulator for classical mechanics called Step

These are some of the actively maintained programs. If you try, you might get many more that are not frequently updated but which might be as useful.

The last bell

The FOSS world offers many options to the field of education and learning. Please remember that we have discussed the applications in isolation; it is up to the teacher and the parent to mix and match the software to suit the learning situation and profile.

Remember the most important thing: Even if some of these software projects are ever suspended or abandoned altogether, the source will always be available in case somebody wants to revive them.

If you are already clued in on the FOSS education scene, this article may not have thrown up any surprises. If you tried even a couple of the programs we discussed, you will appreciate the potential of FOSS in education.

So, while deciding on software for your next generation, is it going to be the pine-scented breeze of freedom—FOSS under the GNU/GPL, or will it be the obfuscating smoke of a restrictive, proprietary EULA (end user licence agreement)? I am not even talking about money here. What price tag will you put on your freedom?

In the next instalment we will discuss how schools can improve their information infrastructure, enabling better access to student and academic data using FOSS solutions. 

References

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By: Gurudutt Talgery

Gurudutt is an open source enthusiast and practitioner with over 20 years of industry experience working with reputed transnational companies. You can contact him at gtalgery.AT.rediffmail.DOT.com.